Yusuf Yiğit PİLAVCI

- yigit.pilavci01@gmail.com
- Website
- Scholar
- Gitlab
- Github

Education and Employement

2025 – 2026	**Postdoctorial Researcher**, IMT Nord Europe Detection of AI-generated images. Supervised by Vincent Itier.
2023 – 2025	**Postdoctorial Researcher**, CRISTAL Lab Design of geometrical priors for inverse problems of bivariate signals. Supervised by Pierre-Antoine Thouvenin, Jérémie Boulanger and Pierre Chainais.
2019 – 2022	**PhD Degree**, Université Grenoble Alpes, GIPSA Lab. Thesis title: *Wilson's Algorithm for Randomized Linear Algebra* Supervised by Nicolas Tremblay, Simon Barthelmé and Pierre-Olivier Amblard. Defended on Nov. 2022.
2017 – 2019	**M.Sc. in Computer Science and Engineering**, Politecnico di Milano Final Grade: 108.0 / 110 Thesis: *Random Spanning Forests, Theory and Applications*
2012 – 2017	**Bachelor's degree in Electrical and Electronics Engineering** in Middle East Technical University. CGPA: 3.79 / 4.00 (10th/375)
2014 – 2017	**Minor degree in Computer Engineering** in Middle East Technical University.

Research Publications

Journal Articles

1. Y. Y. Pilavcı, E. T. Güneyi, C. Cengiz, and E. Vural, "Graph domain adaptation with localized graph signal representations," *Pattern Recognition*, p. 110 628, 2024.
2. Y. Y. Pilavcı, P.-O. Amblard, S. Barthelme, and N. Tremblay, "Graph tikhonov regularization and interpolation via random spanning forests," *IEEE Transactions on Signal and Information Processing over Networks*, 2021.
3. M. Turan, Y. Y. Pilavcı, I. Ganiyusufoglu, H. Araujo, E. Konukoglu, and M. Sitti, "Sparse-then-dense alignment-based 3d map reconstruction method for endoscopic capsule robots," *Machine Vision and Applications*, vol. 29, no. 2, pp. 345–359, 2018.

Conference Proceedings

1. N. Cigercioglu, Y. Pilavci, and H. Guney-Deniz, "Pos1490-hpr can knee osteoarthritis be predicted using machine learning with pedobarographic data," in *Annals of the Rheumatic Diseases*, vol. 84, Elsevier, 2025, pp. 1487–1488.
2. Y. Y. Pilavci, J. Flamant, P.-A. Thouvenin, J. Boulanger, and P. Chainais, "An admm algorithm to restore bivariate signals with joint time and covariance regularization," in *XXXe Colloque Francophone de Traitement du Signal et des Images (GRETSI 25)*, 2025.
3. Y. Y. Pilavci, P. Palud, J. Flamant, P.-A. Thouvenin, J. Boulanger, and P. Chainais, "Time and covariance smoothing for restoration of bivariate signals," in *2025 IEEE Statistical Signal Processing Workshop*, Edinbourgh (Scotland), United Kingdom, Jun. 2025.

4. Y. Y. Pilavcı, J. Boulanger, P.-A. Thouvenin, and P. Chainais, “Denoising bivariate signals via smoothing and polarization priors,” in *2024 32nd European Signal Processing Conference (EUSIPCO)*, IEEE, 2024, pp. 2602–2606.
5. Y. Y. Pilavcı, P.-O. Amblard, S. Barthelme, and N. Tremblay, “Variance Reduction for Inverse Trace Estimation via Random Spanning Forests,” in *GRETSI 2022 - XXVIIIème Colloque Francophone de Traitement du Signal et des Images*, Nancy, France, Sep. 2022.
6. Y. Y. Pilavcı, P.-O. Amblard, S. Barthelmé, and N. Tremblay, “Variance reduction in stochastic methods for large-scale regularised least-squares problems,” in *30th European Signal Processing Conference, (EUSIPCO)*, 2022.
7. Y. Y. Pilavcı, P.-O. Amblard, S. Barthelme, and N. Tremblay, “Smoothing graph signals via random spanning forests,” in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2020, pp. 5630–5634.
8. Y. Y. Pilavcı and N. Farrugia, “Spectral graph wavelet transform as feature extractor for machine learning in neuroimaging,” in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2019, pp. 1140–1144.

Teaching Experience

2023 – 2024 — **IMT Nord**
Security, Authentication and Forensics, Practical work - hour(s) : 3
Compression and error-correcting codes, Group lessons - hour(s) : 3
Compression and error-correcting codes, Tutorial - hour(s) : 9
Signal theory, Tutorial - hour(s) : 12

2024 – 2025 — **IMT Nord**
Signal theory, Tutorial- hour(s) : 20
Tools for data science, Practical work, 3
Centrale Lille
Advanced algorithms and programming,
Tutorial-hours(s): 12 and Practical work-hours(s) : 22

IMT Nord
Signal theory, Tutorial- hour(s) : 12

Skills

Languages	Turkish (Maternal), English (Fluent), French (Intermediate)
Software	Python, Julia, C/C++, Matlab, LaTeX

Industrial Experience

2017 Feb. - May. — **Candidate Engineer, ASELSAN, Ankara/Turkey**
Trained and worked as a software engineer by using C++/C, Unix and Java.

2016 Jun. - Jul. — **Summer Intern, ASELSAN, Ankara/Turkey**
Studied and observed on Real Time Operating Systems on multi-core processors.

Summer Intern, Huawei, Ankara/Turkey
Studied and observed on fiber optical transmission systems and signal modulations.